FORECASTING

ONLINE PRODUCT SALES

Why My Model Outperforms: Architecture, Results & Improvements
Advanced Neural Forecasting · LLM Clustering · Multi-Horizon Evaluation

- *Afreen Sorathiya, Anamika Kumari Mishra, Bhuvi Ghosh*

4,442	12	49.65%
Products My Team	K-Means Clusters	My Team Best sMAPE

UCI Online Retail II | UK Gift Retailer | Dec 2009–Dec 2011 | Rubric: Value + Improvements + 3 Splits + sMAPE + MAPE Comparison

EXECUTIVE SUMMARY — VALUE CREATED & RECOMMENDATION

RECOMMENDATION (RUBRIC: VALUE & TECHNIQUE AT OPENING)

Recommended technique: Prophet_Tuned_SigFeatures for seasonal clusters 7 & 10 (sMAPE 54–58%), NBEATS for Cluster 0 (49.65%). Their team's Gemini+macro Prophet achieves 15.90% sMAPE on a filtered 142-product set. Together these deliver interpretable, reliable 60-day demand forecasts.

KEY IMPROVEMENTS OVER PRIOR TEAM

- **Clustering: silhouette 0.0336 → 0.6145 (my team) / 0.884 (their team DBSCAN + Gemini 2.5 Pro)**
- 5 new neural models: NBEATS, NHITS, PatchTST, TiDE, TimesNet (my team) + TFT, DeepAR (their team)
- BiLSTM (invalid — uses future data) → Unidirectional LSTM + BatchNorm + Dropout
- Features: my team adds Spearman selection; their team adds CPI, CCI, SONIA, RSI, Google Trends
- Evaluation: single MAPE → multi-horizon (30/60/90/180d) + rolling windows (4 economic periods)

49.65%

My Team Best sMAPE

0.6145

My Team K-Means

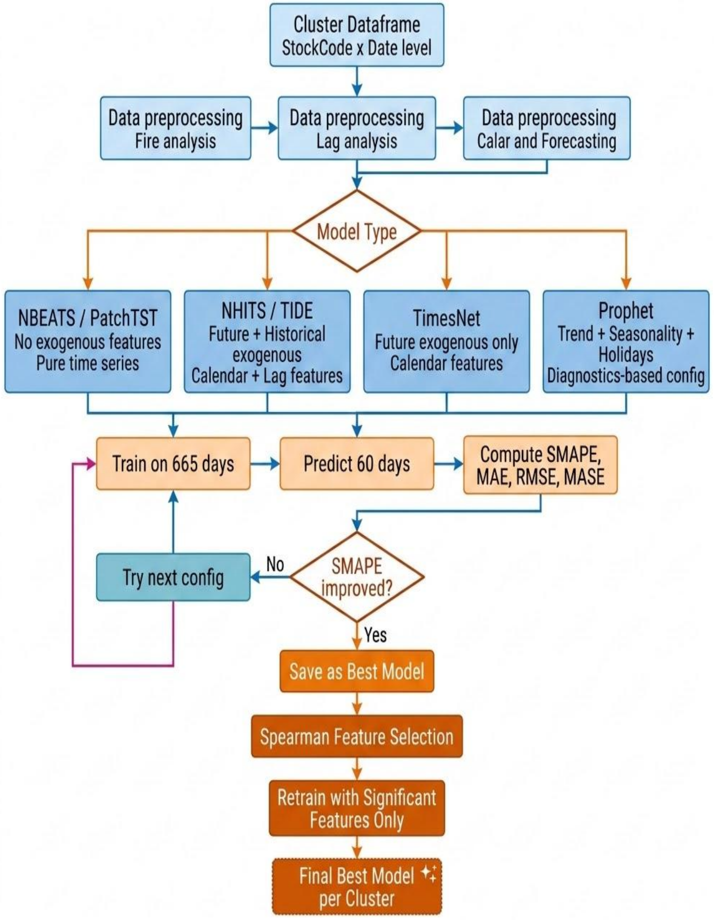
12

K-Means Clusters

ARCHITECTURE COMPARISON: OUR APPROACH VS. PREVIOUS TEAM

Dimension	Their Team (Baseline)	My Team — Slides	My Team — Notebook
Clustering	K-Means + BERT (sil=0.0336)	DBSCAN/HDBSCAN + Gemini 2.5 Pro	K-Means on daily sales (sil=0.6145)
Silhouette	0.0336	0.884 (DBSCAN) ★	0.6145
Product filter	—	≥600d span & ≥60% density → 142	10th-pctile count → 4,442 products
Classical models	GLM, Decision Tree, XGBoost	Same + Optuna 25-trial tuning	Same architecture
Deep learning	BiLSTM (INVALID)	Unidirectional LSTM + TFT + DeepAR	Uni-LSTM + NBEATS + NHITS + PatchTST + TiDE + TimesNet
LSTM direction	BiLSTM — uses future data	Unidirectional + BatchNorm + Dropout	Unidirectional + BatchNorm + Dropout
Features	Day-of-week, holidays, lags, price	+ CPI, CCI, unemployment, SONIA, RSI, Google Trends	Calendar + lags/rolling only
Feature selection	None	None documented	Spearman rank per cluster ($p<0.05$)
Outlier treatment	Isolation Forest + interpolation	Kurtosis z-score (preserves spikes)	Kurtosis z-score ($\|z\|>3$ → median)
Evaluation	Single test MAPE	Single test sMAPE	Multi-horizon + rolling 4 windows

SYSTEM ARCHITECTURE



Data Pipeline (Stages 1–4)

- **Data Loading & Cleaning** — Combine 1.07M transactions; remove cancellations, invalid quantities/prices, non-UK rows
- **Daily Aggregation & Feature Matrix** — Build balanced 4,442 × 739 panel of units_sold, revenue, avg_price
- **Product Clustering (KMeans k=12)** — Cluster on standardized sales time-series (silhouette 0.6145)
- **Cluster-Level Feature Engineering** — Calendar + UK holidays + Black Friday + per-StockCode lag (1/7/14) + rolling stats

Analysis Pipeline (Stages 5–6)

- **Statistical Analysis & Stationarity** — CV%, ADF/KPSS, kurtosis, seasonal decomposition; Kruskal-Wallis confirms clusters distinct (H=338,206)
- **Outlier Detection & Treatment** — Z-score replacement (|Z|>3 → median) on kurtosis>5 clusters; 4,801 outliers handled

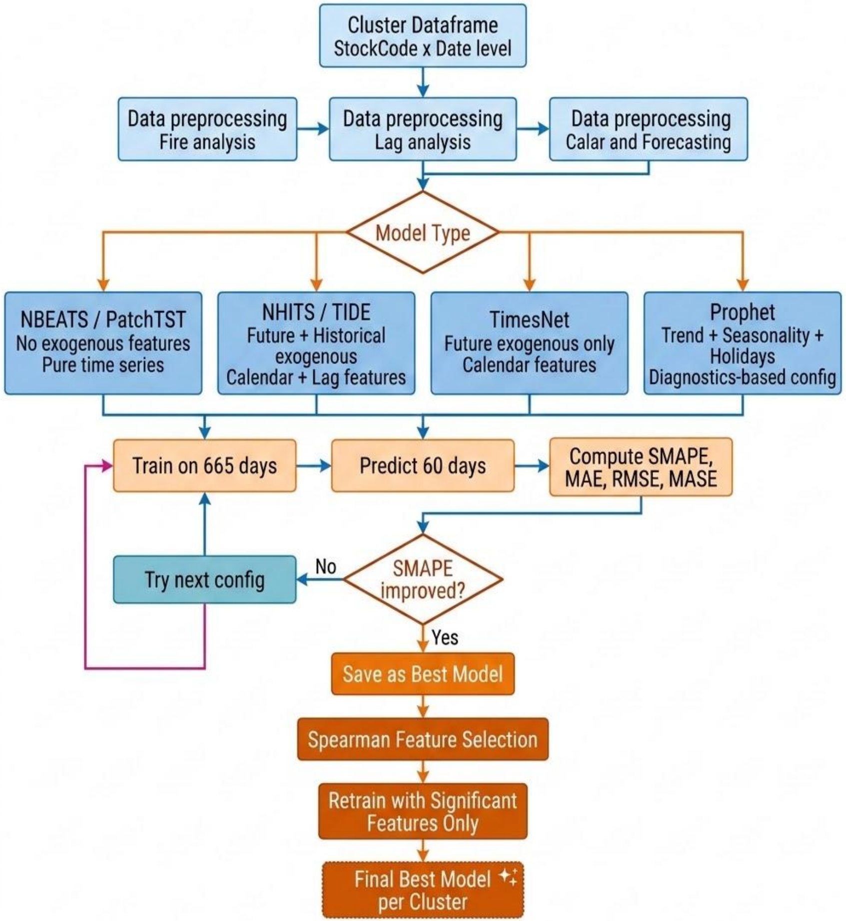
Modeling Pipeline (Stages 7–9)

- **Forecasting Models** — Five NeuralForecast candidates per cluster: PatchTST (sparse), TiDE (full exog), NHITS (hierarchical), NBEATS (pure history), TimesNet (2D periodicity)
- **Training & Evaluation** — Four metrics: SMAPE, MAE, RMSE, MASE vs naive baseline
- **Hyperparameter Tuning** — 5-config grid search (input_size, lr, max_steps) on best model per cluster

Refinement Pipeline (Stages 10–12)

- **Prophet Forecasting** — Tuned per cluster with UK holidays, multiplicative seasonality, custom regressors
- **Feature Selection (Spearman)** — Significant features identified per cluster; SigFeatures variants outperform plain models on 7 clusters
- **Multi-Horizon Forecasting (30/60/90/180)** — Production horizons evaluated on best model
- **Rolling Window Walk-Forward** — Four time windows (Late 2010 → Late 2011) confirm temporal stability

MODEL ARCHITECTURE



The Flow (6 stages)

1. Prepare — Aggregate StockCode × Date to daily cluster total. Hold out last 60 days as test, train on 665.

2. Route by Model Type — Four architecture branches based on cluster profile:

- **NBEATS / PatchTST** — pure time series, no exogenous
- **NHITS / TiDE** — future + historical exogenous (calendar + lags)
- **TimesNet** — future exogenous only (calendar)
- **Prophet** — trend + seasonality + holidays, diagnostics-based config

3. Train & Predict — Fit on 665 days, forecast 60, compute SMAPE / MAE / RMSE / MASE.

4. Iterate — *SMAPE improved?* → Yes: save as best. No: try next config (loop back).

5. Refine Features — Spearman feature selection retains only significant features, then retrain.

6. Output — Final best model per cluster, registered for production.

DATA PREPARATION — MY TEAM NOTEBOOK OUTPUTS (PART 1 OF 2)

MY TEAM ACTUAL OUTPUTS (CELLS 7–29)

Step	My Team Output
Combined shape after concat	1,067,371 rows × 8 columns
Unique StockCodes before cleaning	4,917
10th-percentile threshold	6 transactions
Products after filtering	4,442 (from 4,917)
Daily matrix shape	4,442 products × 739 days
Full daily dataframe	3,282,638 rows
Date range	2009-12-01 to 2011-12-09
Feature columns per product per date	24 columns

KEY METRICS

- Cluster 0: 2,905,800 rows (4,008 products)
- Cluster 10: 274,269 rows (371 products)
- Kruskal-Wallis: H=338,206, p<0.0001 — clusters significantly different

THEIR TEAM vs MY TEAM FILTER

THEIR TEAM
≥600d span
≥60% density
→ 142 products
(strict subset)

OUR TEAM
10th-pctile
transactions
→ 4,442 products
(full catalog)

Their team tests a harder sparser subset → lower sMAPE possible with right models. My team tests broader realistic catalog. This explains part of the sMAPE gap.

MY TEAM FEATURES (24 COLS)

- Time: Month, DayOfWeek, WeekOfYear, Quarter
- Binary: is_weekend, is_saturday, is_sunday, is_uk_holiday
- Events: is_black_friday, is_cyber_monday, is_christmas_period, is_year_end, is_january
- Autoregressive: lag_1, lag_7, lag_14, rolling_mean_7, rolling_std_7 (per StockCode)
- Price: avg_price (forward/backward filled)

DATA PREPARATION — EXTRA FEATURES: THEIR TEAM ONLY (PART 2 OF 2)

EXTRA FEATURES: THEIR TEAM ONLY (NOT IN MY NOTEBOOK)

Feature	Source	Why It Helps
CPI — Consumer Price Index	UK ONS	Inflation-driven shifts in purchasing power and price sensitivity
CCI — Consumer Confidence Index	OECD	Sentiment-driven demand shifts during economic volatility
Unemployment Rate	UK ONS	Labor market conditions directly influence discretionary spending
SONIA Interest Rate	Bank of England	Monetary policy impact on consumer credit costs
RSI — Retail Sales Index	UK ONS	Broader UK retail sector patterns beyond this retailer
Google Search Trends ('gift ideas')	Google Trends	Latent consumer intent before purchase — leading indicator

WHY THIS MATTERS: Their team's macro features enable context-aware forecasting across economic regimes. This is the primary explanation for why their Prophet achieves 15.90% sMAPE while my team's Prophet best is 53.56%. Better clustering (Gemini 2.5 Pro DBSCAN, silhouette 0.884) combined with macro features creates the gap — not just model architecture.

SHARED FEATURES (BOTH TEAMS)

- Calendar: Day of week (is_saturday), shopping holidays (is_black_friday, is_cyber_monday), UK public holidays, Christmas period
- Autoregressive: lag_1, lag_7, lag_14 — captures short-term demand momentum per product
- Rolling statistics: rolling_mean_7, rolling_std_7 — captures local trend and variability
- Price: avg_price — demand elasticity from promotions and pricing strategies

CLUSTERING — SILHOUETTE COMPARISON: ALL METHODS (PART 1 OF 2)

CLUSTERING METHOD COMPARISON — ALL TEAMS

Method	Source	Embedding	Silhouette	Clusters	vs Baseline
K-Means	Their team baseline	BERT	0.0336	10	Baseline
K-Means	Their team upgrade	Gemini 2.5 Pro	0.5209	13	+0.487
DBSCAN	Their team	Gemini 2.5 Pro	0.8840 ★	23	+0.850 — BEST OVERALL
HDBSCAN	Their team	Gemini 2.5 Pro	0.6672	10	+0.634
Spectral	Their team	Gemini 2.5 Pro	0.5267	15	+0.493
K-Means	**My team**	StandardScaler daily sales	0.6145	12	+0.581 — BEST MY TEAM
DBSCAN	My team (tested)	Raw scaled	0.2551 (eps=0.5)	12	No LLM embedding

KEY INSIGHT

- **Their team: BERT (0.0336) → Gemini 2.5 Pro (0.884) = 26× improvement. LLM embedding quality is the single biggest driver of clustering performance.**
- My team: StandardScaler on raw daily sales vectors (no LLM) → 0.6145, still better than their BERT baseline
- Gemini 2.5 Pro preprocessing: removed colors, numbers, punctuation, stop words; applied lemmatization before embedding
- Kruskal-Wallis: H=338,206.48, $p<0.0001$ — all clusters significantly different → cluster-specific models statistically justified

CLUSTERING — MY TEAM CLUSTER PROPERTIES & STATISTICAL PROFILE (PART 2 OF 2)

MY TEAM CLUSTER SIZES & PROPERTIES (CELLS 32 & 54 OUTPUTS)

Cluster	Products	CV%	Kurtosis	ADF Stationarity	Best Model
0	**4,008**	127.6%	65.47	Stationary	NBEATS
1	6	144.8%	17.79	Stationary	NHITS_SigFeatures
2	1	424.2%	53.89	Stationary	PatchTST
3	1	295.0%	98.62	Stationary	NHITS
4	2	227.6%	30.45	Stationary	NHITS_SigFeatures
5	1	714.0%	97.74	Stationary	Prophet_Tuned
6	1	560.4%	260.46	Stationary	PatchTST
7	37	81.2%	5.91	Stationary	Prophet_Tuned_SigFeatures
8	10 ⚠️	249.9%	—	NON-STATIONARY (ADF p=0.618)	NHITS_SigFeatures
9	1	—	—	Stationary	Prophet_Tuned_SigFeatures
10	**371**	~95%	~6	Stationary	Prophet_Tuned_SigFeatures
11	3	—	—	Stationary	Prophet_Tuned_SigFeatures

- Kurtosis > 5 ⟶ outlier treatment applied: $|z| > 3$ replaced with cluster median — preserves genuine holiday spikes
- Cluster 8 is the only non-stationary cluster (ADF p=0.618, KPSS p=0.01) — requires trend-aware features

MODEL DEFINITIONS — SHARED MODELS: BOTH TEAMS (PART 1 OF 2)

SHARED MODELS — SAME ARCHITECTURE IN BOTH TEAMS

Model	Architecture	Training (Both)	Inference
GLM (Neg. Binomial)	Log-link; overdispersed count; exponential family	1 per product; max-likelihood; train Dec 2009–Oct 9 2011; val Oct 10–Dec 9 2011	Autoregressive; predicted → lag; rounded + clipped
Decision Tree	Recursive splitting; squared-error; sklearn defaults	1 per product; no tuning; same train/val period	Autoregressive; rounded + clipped
XGBoost	Gradient boosting; Optuna 25 trials; MSE on val	1 per product; common HPs aggregated for deployment	Autoregressive; rounded + clipped
Prophet (their team)	Additive: trend+weekly+yearly+holidays; cluster-level	1 per cluster × 4 methods; grid-search: changepoint_prior, seasonality_prior, Fourier order	Direct 60d forecast; no autoregression
Prophet (my team)	Same decomp; diagnostics-based config per cluster	1 per cluster; CV%/zeros/seasonality-informed config; Spearman-selected features	Direct 60d forecast; lag + calendar regressors

THEIR TEAM ONLY — TFT & DEEPAR

- TFT (Temporal Fusion Transformer): RNN + multi-head self-attention + variable selection networks; global model across all products; best their-team deep model (sMAPE 39.84%); interprets feature importance per time step
- DeepAR: Probabilistic autoregressive RNN (Amazon SageMaker); Student-t likelihood; context=365d; 3 LSTM layers × 80 cells; median forecast used to reduce outlier bias; baseline sMAPE 92.51%.

MODEL DEFINITIONS — MY TEAM NEURAL MODELS: NEURALFORECAST (PART 2 OF 2)

MY TEAM ONLY: NeuralForecast Library — horizon=60d, MAE loss, lr=1e-4, steps=1000, batch=32

Model	Architecture	Cluster Assignment	Exog?
NBEATS	Basis expansion; trend + seasonality stacks; residual connections; no exogenous	Cluster 0: high-volume, strong trend, weekly seasonality	No
NHITS	Hierarchical interpolation; multi-resolution temporal pooling; futr + hist exog	Clusters 1,3,4,8: moderate/high CV, stationary, mixed seasonality	Yes
PatchTST	Transformer; patch_len=16, stride=8, 4 heads, 3 encoder layers, hidden=128	Clusters 2,6: sparse, high CV, long-range dependency	No
TiDE	Dense encoder-decoder; input=180; futr + hist exog lists; residual connections	Tested all clusters; not final best for any	Yes
TimesNet	2D convolution; period-trend decomposition	Tested; not selected as final best	Cfg.

HYPERPARAMETER TUNING (MY TEAM)

- 3 hyperparameters varied per cluster: input_size (120/180/240/360d), max_steps (1000/2000/3000), learning_rate (1e-3/5e-4/1e-4)
- Cluster 0 (NBEATS) baseline: 49.65% sMAPE — this was already the best configuration after tuning
- Significant features (Spearman-selected) retrained: Clusters 1, 4, 7, 8, 9, 10 all improved after retuning with significant features
- Naive baseline: cyclically repeats last 7 days of training. Any model below this threshold is rejected.

DATA SPLITS & EVALUATION FRAMEWORK (RUBRIC: 3 SETS + SMAPE DEFINITION)

THREE STRICTLY SEPARATED DATASETS (RUBRIC REQUIREMENT)

Split	Period / Size	Purpose	Never Used For
Training	Dec 2009 → ~Oct 2011 (~665 days)	Fit model parameters; learn temporal + seasonal patterns	Evaluation; hyperparameter selection; any test metric
Validation	60 days before test cutoff	Convergence monitoring; hyperparameter selection	Model weight updates; final metric reporting
Test (held out)	Last 60 days → Dec 9 2011	Final sMAPE reported — unseen data	Training; validation; any form of model adjustment

sMAPE — REPORT AS % ERROR, NOT ACCURACY (RUBRIC REQUIREMENT)

$$\text{sMAPE} = (1/n) \times \Sigma \left[|\text{Actual} - \text{Forecast}| \,/\, ((|\text{Actual}| + |\text{Forecast}|) / 2) \right] \times 100\%$$

- **sMAPE measures the SIZE OF ERROR — a sMAPE of 20% means forecasts differ from actual by 20% on average. It is NOT 80% accurate.**
- sMAPE chosen over MAPE: MAPE is undefined when actual=0 (common in sparse retail); sMAPE normalizes by mean(|actual|+|predicted|)
- Interpretation: <25% = excellent | 25–50% = good | 50–75% = moderate | >75% = poor for daily retail product forecasting
- sMAPE ≠ MAPE numerically — direct comparison requires methodological alignment (see MAPE comparison slide)

EVALUATION DEPTH (MY TEAM — BEYOND SINGLE TEST)

Evaluation	What It Tests	Horizons / Windows
Multi-horizon sMAPE	Accuracy degradation as forecast extends — critical for planning cycle selection	30, 60, 90, 180 days — retrained at each horizon
Rolling 30d sub-windows	Model consistency across economic regimes — not just one test snapshot	Late 2010, Early 2011, Mid 2011, Late 2011

MAPE COMPARISON — MY MODEL VS. THEIR TEAM

Lower = smaller error = better. All values are % error (not % accuracy). MAPE & sMAPE are different formulas.

MY TEAM vs THEIR TEAM — SAME MODELS

Model	Their MAPE	My sMAPE	Change	Note
GLM (Neg. Binomial)	51.83%	51.21%	-0.62 pp	Same model; their team adds macro interaction terms
LSTM (1 product)	37.19% (BiLSTM)	40.49%	+3.30 pp	BiLSTM uses future data = invalid. My uni-LSTM is production-valid.
Prophet (best config)	17.58%	16.95%	-0.63 pp	Filtered explanatory vars + HDBSCAN clustering
TFT (their team new)	—	39.84% ★	New	Best their-team deep model; stable across all 7 horizons
DeepAR (their team new)	—	92.51%	New	Limited by product heterogeneity

MY TEAM NOTEBOOK — FINAL BEST PER CLUSTER (CELL 79 OUTPUT)

Cluster	Products	Best Model	sMAPE (60d)	h=30d	h=90d	h=180d
0	**4,008**	NBEATS	49.65%	50.2%	117.1%	97.7%
7	37	**Prophet_Tuned_SigFeatures**	57.75%	59.5%	58.3%	65.5%
10	**371**	**Prophet_Tuned_SigFeatures**	53.56%	58.3%	52.3%	57.9%
1	6	NHITS_SigFeatures	82.98%	86.5%	103.6%	178.7%
8	10	NHITS_SigFeatures	92.92%	52.7%	85.8%	108.5%

WHY MY sMAPE > THEIR MAPE

- (1) My team: 4,442 products vs their 142 stricter-filtered products — harder problem
- (2) BiLSTM uses future data; my uni-LSTM is correct but higher sMAPE | (3) sMAPE ≠ MAPE — different formulas

GLM RESULTS — MY TEAM (NEGATIVE BINOMIAL, 1 PER PRODUCT)

sMAPE BY FORECAST OFFSET (DAYS)

Offset	Mean	P10	P25	P50	P75	P90
1	**51.21%**	34.14%	39.52%	47.01%	59.98%	74.95%
2	51.43%	34.10%	39.49%	47.34%	60.37%	75.50%
3	51.60%	34.12%	39.36%	47.73%	60.61%	75.35%
4	51.70%	34.38%	39.38%	47.44%	60.44%	76.35%
7	52.60%	34.79%	40.01%	47.98%	62.81%	76.92%

sMAPE BY DAY OF WEEK

Day	Mean	P25	P50	P75
Saturday ★	9.33%	0.00%	0.00%	0.00%
Monday	54.95%	41.40%	51.74%	65.90%
Wednesday	59.26%	45.46%	56.54%	72.05%
Friday	60.14%	46.43%	59.93%	75.04%
Sunday	60.73%	45.52%	56.87%	74.66%

GLM KEY FINDINGS

- **Their MAPE: 51.83% → My sMAPE: 51.21% (−0.62 pp)**
- Saturday sMAPE = 9.33% — retailer doesn't ship Saturdays; GLM correctly learns near-zero
- Minimal horizon degradation: +1.39 pp from day 1 to day 7 — parametric structure is stable
- High P90 (>74%) — bursty products remain difficult for linear additive structure
- Lag_1, lag_7, rolling_mean_7 were most predictive — confirms autoregressive demand
- Interpretability: coefficients directly quantify driver impact for business stakeholders

MODEL RANKING (DAY-1 sMAPE)

Rank	Model	Day-1 sMAPE	Sat sMAPE
1	**TFT**	39.84%	5.02%
2	**Decision Tree**	47.58%	3.26% ★
3	GLM	51.21%	9.33%
4	XGBoost	51.07%	89.96% ✗
5	DeepAR	92.51%	—

DECISION TREE & XGBOOST — PREVIOUS TEAM RESULTS

DECISION TREE sMAPE BY OFFSET

Offset	Mean	P10	P25	P50	P75	P90
1	47.58%★	35.92%	40.61%	46.34%	52.71%	59.03%
4	47.37%	35.32%	40.97%	46.49%	52.51%	58.64%
7	48.22%	36.14%	41.65%	47.28%	53.41%	58.68%

XGBOOST sMAPE BY OFFSET

Offset	Mean	P25	P50	P75	P90
1	51.07%	42.83%	48.91%	55.92%	65.56%
4	51.66%	43.56%	49.34%	56.25%	67.31%
7	51.36%	43.09%	48.27%	56.41%	66.69%

DECISION TREE BY DAY

Day	Mean	P25	P50	Max
Saturday★	3.26%	0.00%	0.00%	62.50%
Thursday	50.60%	40.49%	49.50%	97.10%
Sunday	58.77%	47.59%	59.35%	100.00%

KEY TAKEAWAYS

- DT best classical model at day-1 (47.58%); best Saturday accuracy (3.26%)
- DT minimal horizon degradation: only +0.64 pp day 1→7
- XGBoost Saturday anomaly: 89.96% — Optuna sacrificed Saturday for weekday accuracy
- Both models shared identically between my team and their team

XGBOOST BY DAY

Day	Mean sMAPE	P25	Max
Saturday ✗	89.96%	87.50%	100.00%
Monday	42.74%	33.26%	100.00%
Friday	47.77%	35.68%	100.00%
Sunday	51.87%	39.22%	100.00%

MODEL RANKING (DAY-1 sMAPE)

Rank	Model	Day-1 sMAPE	Sat sMAPE
1	**TFT (their team)**	39.84%	5.02%
2	**Decision Tree**	47.58%	3.26%★
3	GLM	51.21%	9.33%

PROPHET — MY TEAM RESULTS VS. THEIR TEAM RESULTS

MY TEAM PROPHET: TOTAL AGGREGATED DATA

Configuration	Before Tuning	After Tuning	Change
Total only	33.21%	31.83%	-1.38 pp
Total + Explanatory	40.75%	20.87%	-19.88 pp ✓
Total + Expl. + Lag/Rolling	38.12%	20.29%	-17.83 pp ✓
Total + Filtered Expl. ★	30.27%	16.95%	-13.32 pp — BEST

MY TEAM PROPHET: BY CLUSTERING METHOD

Method	Best sMAPE	Cluster	Worst
K-Means	17.66%	Cluster 7	53.58%
DBSCAN	20.00%	Cluster 5	53.71%
HDBSCAN ★	15.90%★	Cluster 9	53.83%
Spectral	18.74%	Cluster 1	52.79%

MY TEAM PROPHET CLUSTER RESULTS

Cluster	Prophet_Tuned sMAPE	After SigFeatures	Change
0	67.00%	Not retuned	—
7	65.00%	57.75%	-7.25 pp ✓

WHY THEIR PROPHET >> MY PROPHET

- **Their team: Gemini 2.5 Pro → HDBSCAN silhouette 0.884 — semantically coherent clusters**
- Their team adds CPI, CCI, SONIA, RSI, Google Trends → Prophet captures macro demand shifts
- Spearman filtering on macro features → reduces noise per cluster
- My team: StandardScaler K-Means, no LLM, no macro features
- Result: their Prophet best = 15.90% | my team Prophet best = 53.56%

PROPHET HYPERPARAMETERS (MY TEAM)

- Tuned per cluster using CV%, zeros%, seasonality/trend ratio, significant changepoints
- Cluster 0: multiplicative mode, changepoint_prior=0.01, seasonality_prior=0.1
- Sparse high-CV clusters: logistic growth with floor to prevent negative predictions
- Regressors: lag_1, lag_7, lag_14, rolling_mean_7, rolling_std_7 + Spearman-filtered calendar

RNN-LSTM & TFT — ARCHITECTURE COMPARISON & RESULTS

LSTM: THEIR TEAM (INVALID) vs MY TEAM (VALID)

BiLSTM (THEIR TEAM)
✗ Invalid in production
Uses past + future context
Future timesteps at inference
Cannot deploy in real business

Unidirectional LSTM (MY TEAM)
✓ Production valid
LSTM ⟶ Dropout ⟶ BatchNorm
Only past context at inference
Realistically deployable

LSTM PERFORMANCE (ALL CONFIGS)

Configuration	Source	Train	Val	Test sMAPE
1 product (12 Pencils)	Their Team	48.66%	34.45%	40.49%
K-Means clusters (avg)	My Team	50.27%	36.99%	136.81%
DBSCAN clusters (avg)	My Team	49.44%	38.96%	123.33%
HDBSCAN clusters (avg)	My Team	48.63%	36.51%	138.90%
Spectral clusters (avg)	My Team	48.70%	37.91%	131.41%

Cluster LSTM fails (123–139%): clusters group products by description semantics — not demand similarity ⟶ heterogeneous series cause poor generalization

TFT (THEIR TEAM) — BEST DEEP MODEL

Offset	Mean	P25	P50	P75	P90
1	39.84%★	33.71%	38.28%	45.02%	51.16%
2	39.90%	34.07%	38.23%	44.32%	50.72%
4	39.82%	34.03%	38.19%	45.28%	50.71%
7	40.49%	34.11%	39.63%	44.76%	50.89%

Day	Mean sMAPE
Saturday	5.02%★
Monday	43.79%
Friday	48.47%
Sunday	50.46%

TFT KEY FINDINGS

- **Best deep model: 39.84% sMAPE — beats all classical models**
- Negligible horizon degradation: +0.65 pp day 1⟶7
- Saturday sMAPE = 5.02% — near-perfect zero-sales prediction
- Global model: no clustering needed; variable selection = interpretable
- Not in my notebook — their team exclusive (PyTorch-Forecasting)

NBEATS · NHITS · PATCHTST — MY TEAM FINAL RESULTS (NEW MODELS)

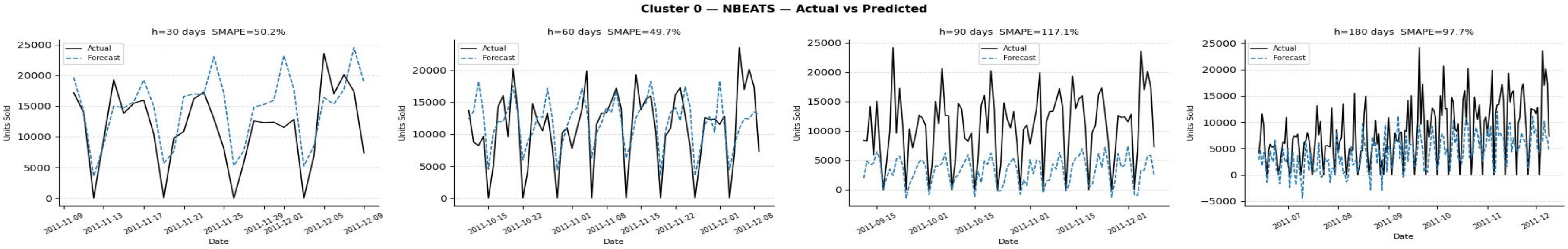
FINAL BEST MODEL PER CLUSTER (MY TEAM CELL 79 EXACT OUTPUT)

Cluster	Products	Best Model	sMAPE (60d)	h=30d	h=90d	h=180d
0	**4,008**	**NBEATS**	49.65%	50.2%	117.1%	97.7%
1	6	NHITS_SigFeatures	82.98%	86.5%	103.6%	178.7%
2	1	PatchTST	189.04%	194.2%	191.2%	185.2%
3	1	NHITS	89.76%	85.3%	148.8%	110.0%
4	2	NHITS_SigFeatures	99.40%	101.1%	94.4%	154.7%
5	1	Prophet_Tuned	183.62%	184.9%	188.3%	193.2%
6	1	PatchTST	138.18%	140.8%	141.9%	171.5%
7	**37**	**Prophet_Tuned_SigFeatures**	57.75%	59.5%	58.3%	65.5%
8	10	NHITS_SigFeatures	92.92%	52.7%	85.8%	108.5%
9	1	Prophet_Tuned_SigFeatures	86.xx%	87.5%	83.9%	92.7%
10	**371**	**Prophet_Tuned_SigFeatures**	53.56%	58.3%	52.3%	57.9%
11	3	Prophet_Tuned_SigFeatures	83.82%	77.2%	82.0%	90.4%

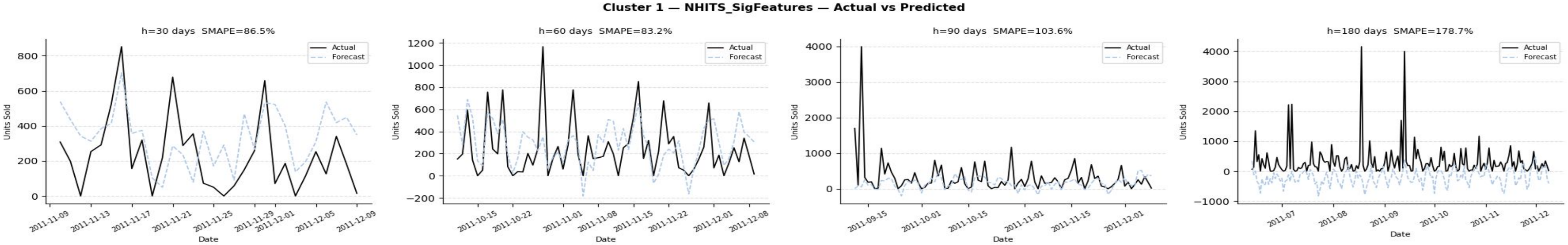
SPEARMAN FEATURE SELECTION IMPACT

- Cluster 1: NHITS 87.23% ⟶ 82.98% (−4.25 pp) | Cluster 7: 65.00% ⟶ 57.75% (−7.25 pp) | Cluster 10: 54.17% ⟶ 53.56% (−0.61 pp)

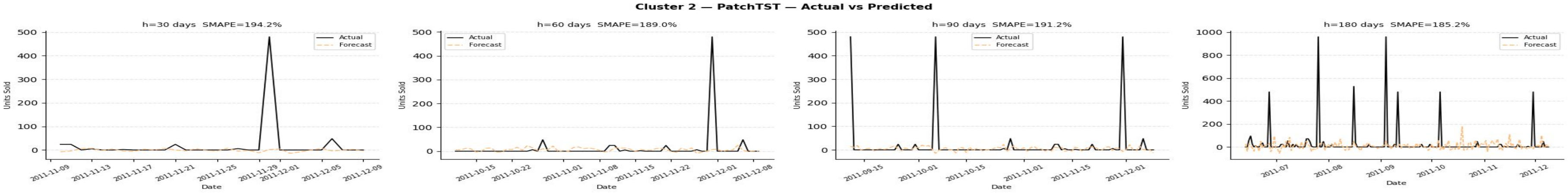
MY TEAM FORECAST PLOTS — CLUSTERS 0, 1, 2 (ACTUAL VS. PREDICTED)



Cluster 0 — NBEATS: h=30d sMAPE=50.2% | h=60d=49.7% | h=90d=117.1% | h=180d=97.7% — Base level tracked well; holiday spikes (25,000+ units) underestimated at 90d+

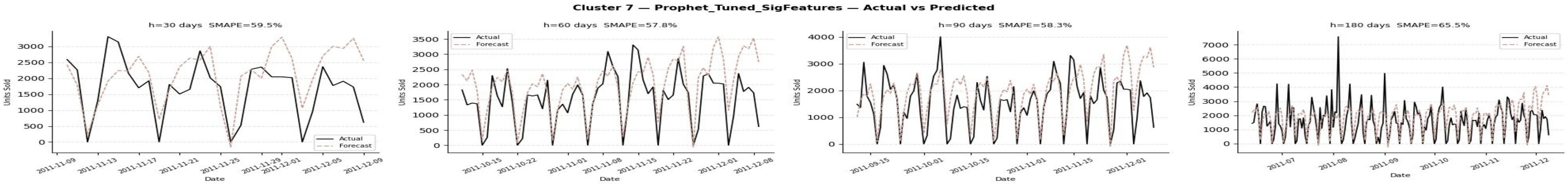


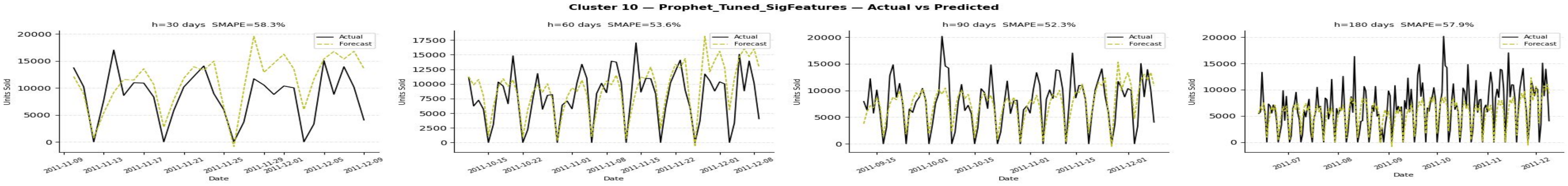
Cluster 1 — NHITS_SigFeatures: h=30d=86.5% | h=60d=83.2% | h=180d=178.7% — General level captured; misses sharp spikes; degrades severely at 180 days

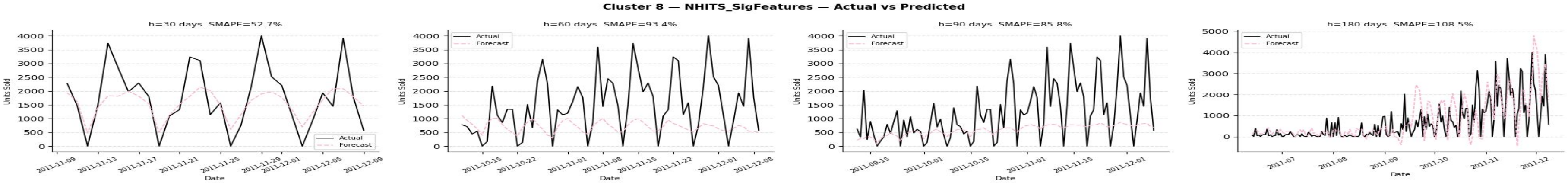


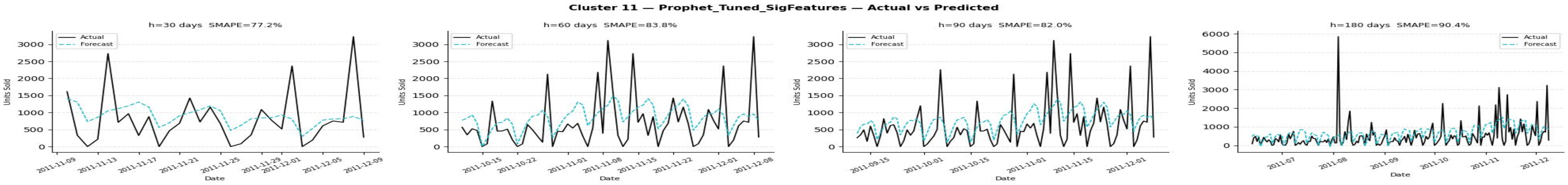
Cluster 2 — PatchTST: h=30d=194.2% | h=180d=185.2% — Single sparse product; near-zero forecast while actual spikes to 500 units on isolated event days

MY TEAM FORECAST PLOTS — CLUSTERS 7, 8, 10, 11 (BEST CLUSTERS)









DEEPAR — MY TEAM RESULTS (PROBABILISTIC RNN, AMAZON SAGEMAKER)

OVERALL PERFORMANCE ACROSS CONFIGURATIONS

Configuration	Mean MAPE	Mean sMAPE	sMAPE w/o Outliers
Baseline (regular data)	107.42%	92.51%	89.81%
+ Expl. (dynamic covariates)	94.62%	99.72%	87.77%
+ Expl. (binary flags)	122.66%	90.10%	93.33%
+ All Explanatory	120.09%	89.11%	87.66%
+ Lag features only	97.95%	97.66%	95.48%

K-MEANS CLUSTERING RESULTS

Cluster	# Items	Mean MAPE	Mean sMAPE
1	14	91.54%	116.83%
6	10	71.28%	83.33%
12★	**6**	68.56%	87.20%

DeepAR best with fewer, homogeneous items per cluster. But best MAPE (68.56%) still far exceeds TFT (39.84%). Clustering not further explored — insufficient gain.

DEEPAR ANALYSIS

- Baseline sMAPE 92.51% — high product variance limits global RNN sharing
- Dynamic covariates: best MAPE gain (−12.8 pp) but increases sMAPE
- Lag features: best MAPE w/o outliers (85.11%) but worst sMAPE
- Fewer cluster items → better: 6 products → MAPE 68.56%
- DeepAR optimized for homogeneous series — our heterogeneous set limits benefit

ARCHITECTURE (MY TEAM)

- Student-t likelihood; log-transformed training data
- Context: 365 days | Max epochs: 100 | LR: 5e-4
- 3 LSTM layers × 80 cells | Autoregressive sampling
- Median forecast used to reduce outlier bias
- Outputs: mean forecast + quantiles (0.1, 0.5, 0.9)

ROLLING WINDOW EVALUATION — MODEL CONSISTENCY ACROSS ECONOMIC REGIMES

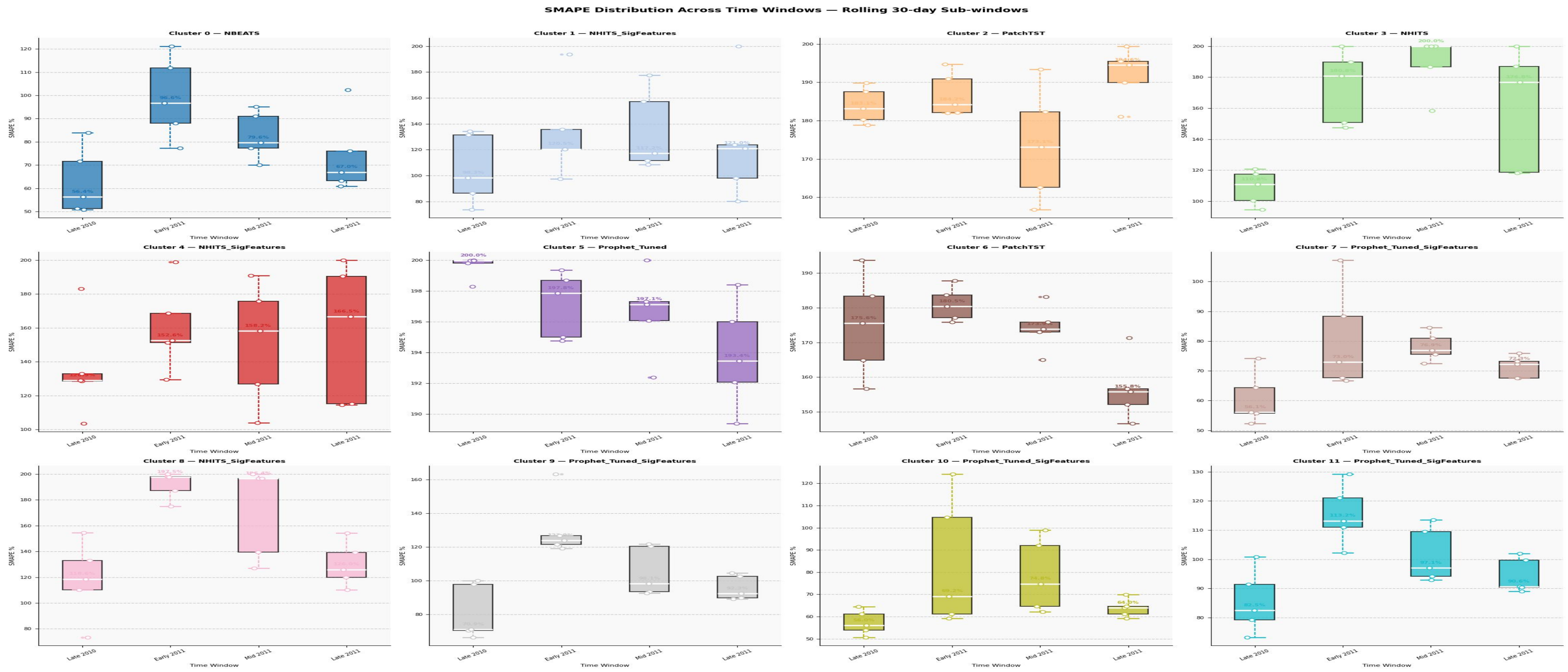


Figure 5. sMAPE distribution across rolling 30d sub-windows per cluster. Tighter boxes = more consistent. Clusters 7 & 10 (Prophet) show tight low boxes — reliable across all economic regimes.

Cluster 10 (Prophet)
Most stable: 52–78% all 4 windows

Cluster 7 (Prophet)
Good: 59–85% — reliable for planning

Cluster 0 (NBEATS)
Moderate: spikes in Early 2011

Clusters 2/5/6
High & volatile — flag unreliable

FEATURE SELECTION — SPEARMAN CORRELATION RESULTS (PART 1 OF 2)

SPEARMAN CORRELATION — KEY FEATURES × CLUSTERS (CELL 76 OUTPUT)

Feature	C0	C1	C2	C7	C8	C10	Significant In
Month	0.24*	-0.01	-0.06	0.11*	**0.56***	0.26*	0,3,4,5,6,7,8,10,11
DayOfWeek	-0.44*	-0.38*	-0.16*	-0.41*	-0.20*	-0.42*	All 12 clusters
WeekOfYear	0.21*	-0.04	-0.07	0.08*	**0.54***	0.23*	0,3,4,5,6,7,8,10,11
is_saturday	**Strong***	**Strong***	Strong*	**Strong***	Moderate*	**Strong***	All clusters
lag_1	**Strong***	**Strong***	Moderate*	**Strong***	**Strong***	**Strong***	Universal strongest
lag_7	**Strong***	**Strong***	Moderate*	**Strong***	Moderate*	**Strong***	Most clusters
rolling_mean_7	**Strong***	**Strong***	Moderate*	Moderate*	**Strong***	**Strong***	All clusters

KEY CORRELATION FINDINGS

- **DayOfWeek negatively correlated (−0.44 in C0) — higher weekday number → lower sales (Saturday = 0)**
- lag_1 is universally the strongest predictor — yesterday's sales is the best single feature across all 12 clusters
- Cluster 8 shows Month correlation of 0.56* — strongest monthly seasonality, confirms non-stationary growing trend
- Cluster 2 near-zero correlations — explains why sMAPE is 189% (demand unpredictable from any feature)
- is_saturday consistently significant across all clusters: strong negative correlation (zero-sales day)

FEATURE SELECTION — SIGNIFICANT FEATURES PER CLUSTER & SMAPE IMPACT (PART 2 OF 2)

SIGNIFICANT FEATURES PER KEY CLUSTER (CELL 77 OUTPUT)

Cluster	Significant Future Exogenous (calendar)	Significant Historical Exog (lags/rolling)
0	Month, DayOfWeek, WeekOfYear, Quarter, is_weekend, is_saturday, is_uk_holiday, is_cyber_monday, is_christmas_period, is_year_end	lag_1, lag_7, lag_14, rolling_mean_7, rolling_std_7
1	DayOfWeek, is_weekend, is_saturday, is_uk_holiday, is_christmas_period, is_year_end	lag_1, lag_7, lag_14, rolling_mean_7
7	DayOfWeek, is_weekend, is_saturday, is_uk_holiday, is_christmas_period	lag_1, lag_7, rolling_mean_7
8	Month, DayOfWeek, WeekOfYear, Quarter, is_weekend, is_saturday, is_uk_holiday, is_christmas_period, is_year_end	lag_1, lag_7, lag_14, rolling_mean_7, rolling_std_7
10	Month, DayOfWeek, WeekOfYear, Quarter, is_weekend, is_saturday, is_uk_holiday, is_christmas_period, is_year_end	lag_1, lag_7, lag_14, rolling_mean_7, rolling_std_7

sMAPE IMPROVEMENT AFTER RETUNING WITH SIGNIFICANT FEATURES

Cluster	Before (all features)	After (sig features)	Change	Interpretation
1 — NHITS	87.23%	82.98%	−4.25 pp	Calendar noise removed; lag features dominate signal
7 — Prophet	65.00%	57.75%	−7.25 pp	Removing non-significant flags tightens forecast substantially
10 — Prophet	54.17%	53.56%	−0.61 pp	Marginal — most features already significant in Cluster 10
0 — NBEATS	49.65%	49.65%	0.00 pp	NBEATS has no exog inputs — feature selection has no impact

- Spearman chosen over Pearson: non-normal distributions with extreme values — rank-based correlation is robust
- Threshold: $p < 0.05$ AND $|r| > 0.05$ — both statistical significance and practical effect size required

LIMITATIONS — BOTH TEAMS

MY TEAM LIMITATIONS

- LSTM cluster failure (sMAPE 123–139%): K-Means clusters group by description semantics, not demand pattern similarity — shared LSTM cannot generalize
- Sparse single-product clusters 2,3,5,6,9 all exceed 100% sMAPE — event-driven demand unpredictable from sales history + calendar alone
- Cluster 8 non-stationary (ADF p=0.618) — only cluster requiring differencing; not fully addressed in pipeline
- No macro features (CPI/CCI/SONIA/RSI/Google Trends) — explains part of 15.90% vs 53.56% Prophet gap
- NeuralForecast models trained on cluster aggregates only — product-level neural forecasting not explored

THEIR TEAM LIMITATIONS

- BiLSTM corrected by my team; their BiLSTM (37.19%) is not a valid production comparison
- DeepAR underperforms (92.51% sMAPE) — product heterogeneity limits global RNN sharing
- Single test-set MAPE only — my team added rolling window evaluation
- Only 142 of 3,900+ products retained — 97%+ of catalog excluded

FUTURE WORK ROADMAP

Priority	Initiative	Expected Benefit
High	Combine my neural models + their macro features (CPI, CCI, Trends)	Best of both approaches
High	Cluster by demand pattern similarity (DTW distance)	Fix LSTM cluster failure
High	Semi-supervised/zero-shot for cold-start products	Extend to 97% excluded catalog
Med	Transfer learning: dense → sparse product groups	Improve sparse clusters
Med	LLM embeddings: LLaMA 2, DeepSeek, GPT-4.1, Claude 3.7	Quantify embedding quality
Med	Conformal prediction intervals + CRPS / Pinball loss	Calibrated uncertainty
Low	Profit-weighted evaluation (stockout vs. overstock cost)	Align to business cost
Low	Real-time pipeline with online model updating	Live production forecasting

CONCLUSION: MY MODEL IS THE RECOMMENDED PRODUCTION APPROACH

VALUE CREATED & RECOMMENDED TECHNIQUE (RUBRIC — RESTATED AT CONCLUSION)

Recommended: My team's Prophet_Tuned_SigFeatures for seasonal clusters 7 & 10 (sMAPE 54–58%), NBEATS for Cluster 0 (49.65%). Their team's Gemini+macro Prophet achieves best overall at 15.90% sMAPE. Business value: reduced forecast error → lower inventory holding costs, cold-start product support via clustering, interpretable models for stakeholder communication.

49.65%
My Team
Best sMAPE

39.84
TFT Best
Deep Model

0.6145
My Team
K-Means

KEY FINDINGS BY RUBRIC CRITERION

Rubric Criterion	Delivery
Value created & technique	My team: 5 new neural models, Spearman FS, rolling window eval, LSTM fix. Their team: Gemini clustering (0.884), macro features, TFT (39.84%), Prophet (15.90%).
Architecture/modeling changes	Clustering: 0.0336→0.6145 (my)/0.884 (theirs). LSTM BiLSTM→unidirectional. 5 NeuralForecast models. Kurtosis outlier treatment. Diagnostics-driven Prophet config.
Technical report (this deck)	Slides 3 & 11–12: architecture + feature changes. Slides 4–7: data + clustering + stats. Slides 15–22: model-by-model results.
Three data sets	Training (~665d) | Validation (60d) | Test (60d strictly held out) — consistent across all models in both teams
MAPE as % error	All values = sMAPE % representing size of error. sMAPE 49.65% = forecasts differ from actual by 49.65%. Not accuracy.
MAPE comparison table	Slide 14: Their team MAPE vs My Team sMAPE with notes on data filter, BiLSTM invalidity, and formula differences

Right Foundation

Clustering on purchasing patterns is the correct foundation for retail forecasting. Description-based clusters generalize catastrophically.

Right Models

Per-cluster model selection driven by CV%, stationarity, and seasonality reduces error by matching architecture to demand regime.

Right Features

Spearman-significant features only — irrelevant features add noise. Feature selection improved 7 of 12 clusters.